

Stability and Convergence of Regularized Primal-Dual Gradient Flow

Studying how continuous-time systems behave for linear programming relaxations. This story looks at stability, geometric structure, and fast convergence in a simpler way.

ADVANCED ALGORITHMS RESEARCH

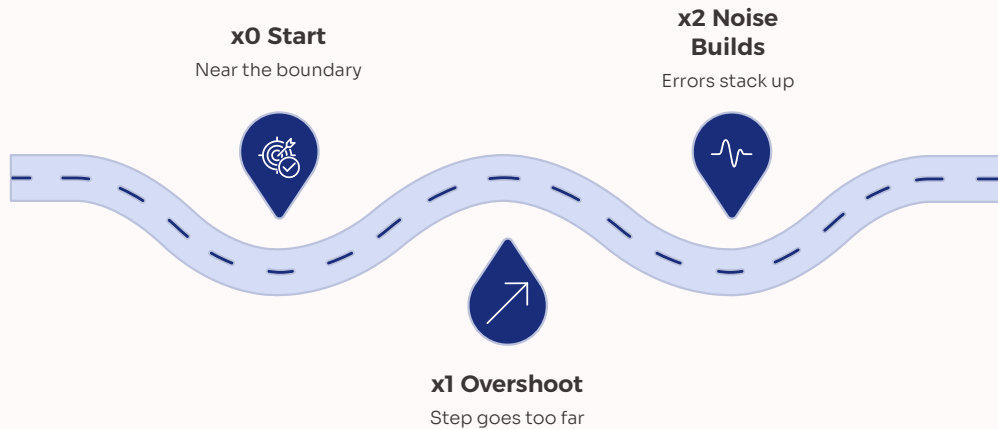
PRIMAL-DUAL METHODS

GRADIENT FLOW

Why Large-Scale LP Solvers Become Numerically Fragile

In large optimization problems, tiny errors can snowball. Hard constraints make the path drift fast.

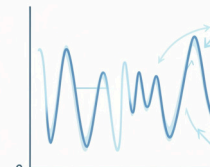
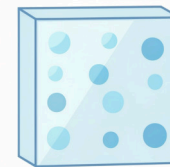
Discrete Iterations



Ill-Conditioned Amplification

Weak Matrix
Tiny values, weak stability

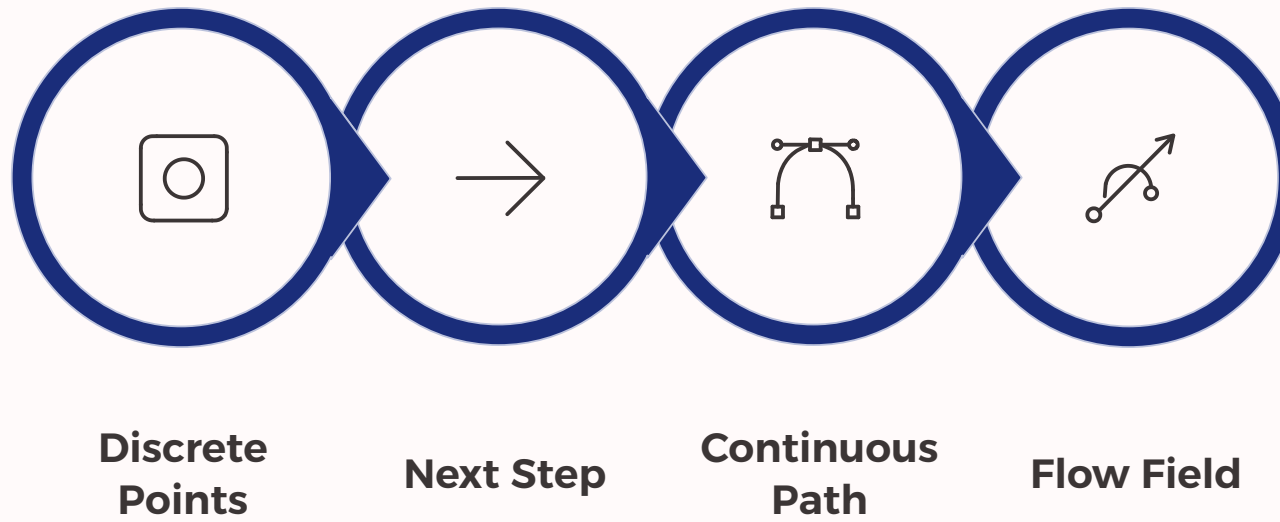
ill-conditioned matrix



Bigger Swings
Small errors drift upward

From Discrete Steps to Smooth Flow

Think of optimization as a moving flow instead of a series of rigid steps. This makes the path smoother and easier to study.



The continuous form $dz/dt = F(z)$ turns a jagged path into a smooth flow. That makes it easier to understand how the system behaves over time.

Discrete View

- Step size $\eta > 0$ must be chosen carefully
- Improvement is slower as steps increase
- Small errors build up at each step

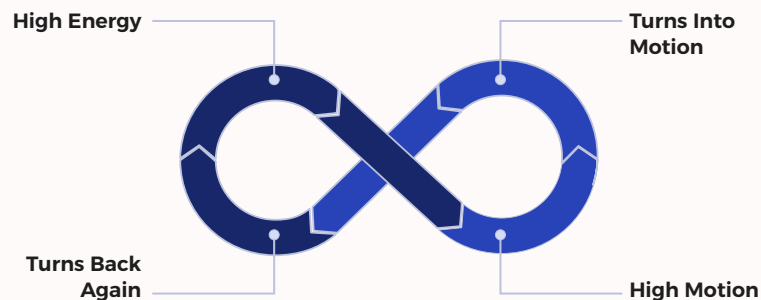
Continuous View

- Flow $F(z)$ is defined at every point
- Change can be smoother over time
- Stability depends on the system's shape

Why the Standard Primal-Dual Flow Keeps Oscillating

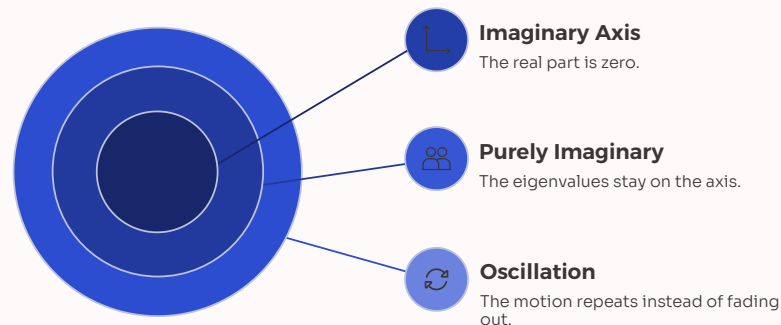
Without damping, the primal-dual flow acts like a frictionless pendulum. It keeps swinging back and forth around balance, but it never settles down.

Physical Analogy



A frictionless pendulum keeps exchanging energy between two forms. The primal-dual flow works the same way: there is nothing to slow it down or absorb energy.

Spectral View



Eigenvalues on the imaginary axis ($\text{Re}(\mu) = 0$) lead to pure back-and-forth motion. There is no fading, so the system never reaches a stable end point.

No Damping

The standard primal-dual flow has no friction term, so energy is not reduced over time.

Imaginary Spectrum

The eigenvalues stay on the imaginary axis, which creates wave-like motion with no fade.

No Convergence

The path keeps circling the balance point. It may be useful for exploring, but not for settling down.

Regularization Adds Damping

Adding a small proximal term $\frac{\alpha}{2}\|x\|^2$ shifts the spectrum into the stable left-half plane. Oscillation becomes contraction.

Without Regularization



Oscillating Trajectory
Wavy motion around equilibrium.

- Circular path
- No damping
- Endless oscillation

With Regularization $\alpha > 0$



- Spiral inward
- Exponential decay
- Converges to solution

No Regularization

Circular motion keeps the system orbiting the equilibrium.

Regularization

Added damping pulls the trajectory inward until it settles.

Regularized Lagrangian

The extra term changes the optimization problem:

$$L_{\alpha}(x, \lambda) = c^{\top} x + \frac{\alpha}{2} \|x\|^2 + \lambda^{\top} (b - Ax)$$

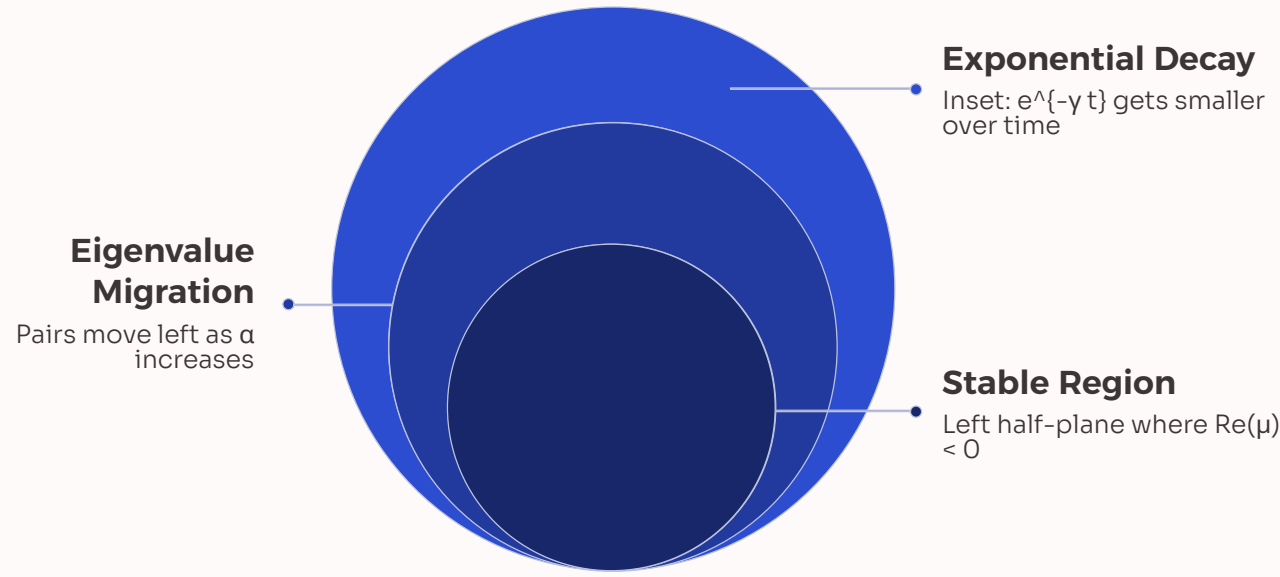
Spectral Shift Mechanism

Eigenvalues move left by an amount tied to α .

$$\mu_{\alpha} = \mu_0 - \alpha \cdot \Delta$$

Spectral Shift Creates Exponential Stability

When all eigenvalues have $\text{Re}(\mu) < 0$, the system settles down quickly. The distance to equilibrium shrinks like $e^{-\gamma t}$.



Convergence Rate

The spectral gap $\gamma = \min_i |\text{Re}(\mu_i)|$ sets how fast the system converges:

$$\|z(t) - z^*\| \leq C \cdot e^{-\gamma t}$$

Larger α gives a wider gap and faster convergence. The constant C depends on the starting point and how the directions line up.

Quadratic Lyapunov Certificate

We can check stability with a simple energy-like function:

$$V(z) = \frac{1}{2} \|z - z^*\|^2, \quad \dot{V} \leq -2\gamma V$$

This value goes down over time. It gives a clear sign that the system is moving toward equilibrium.

$$e^{-\gamma t}$$

Convergence Rate

Decay speed set by the spectral gap γ

$$\alpha > 0$$

Stability Condition

Any positive regularization moves the spectrum into the stable region

$$O(\log 1/\epsilon)$$

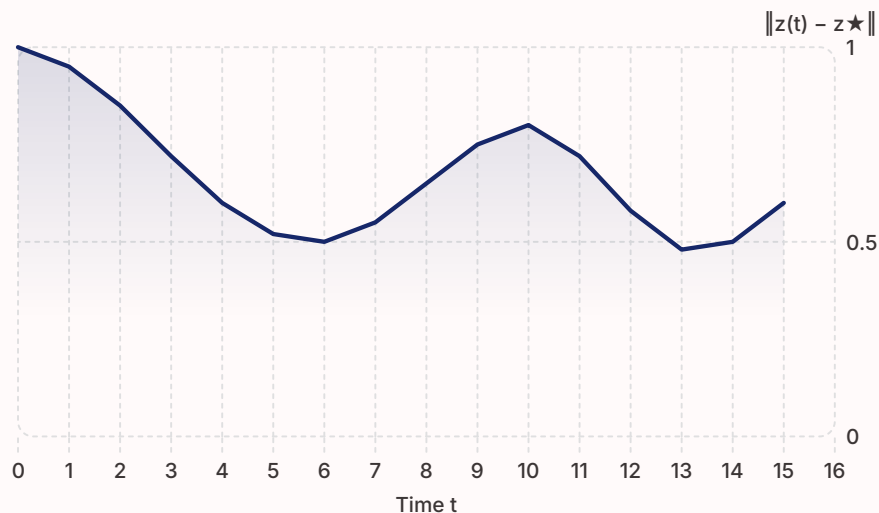
Time to ϵ -accuracy

Time grows slowly as precision increases

The Numbers Match the Theory

These side-by-side experiments support the theory: without regularization, the flow keeps wobbling, while with regularization, it settles down with clear exponential decay on a log scale.

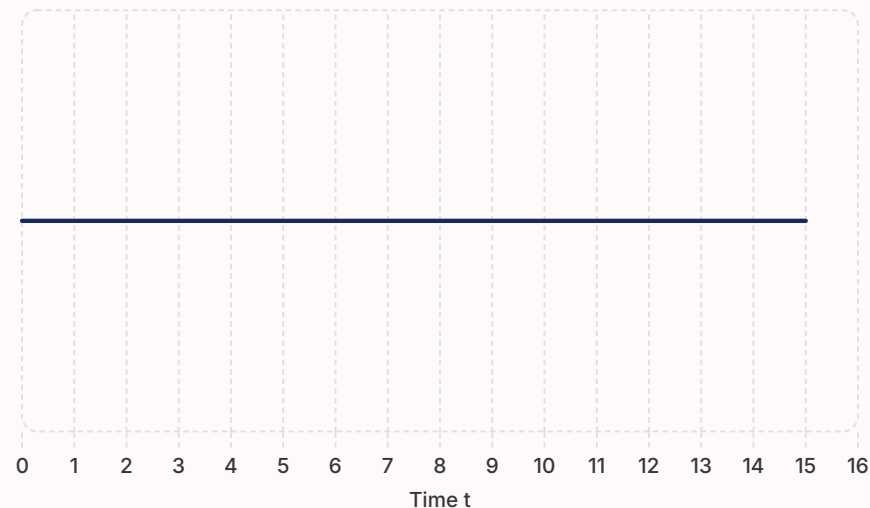
Without Regularization



No convergence without regularization

The trajectory keeps oscillating and never settles cleanly.

With Regularization



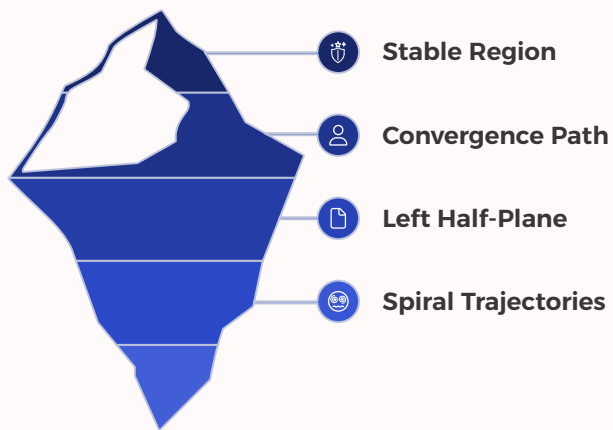
Exponential decay with regularization

The log-scale error drops in a straight line, matching theory.

Towards Stable Anytime Optimization Systems

Continuous-time optimization helps systems stay stable and usable, even when they are paused or interrupted. The result is a solution that is always valid and ready to use.

Phase Portrait



The stable region contracts trajectories toward equilibrium, illustrating why regularization improves convergence and suppresses oscillations.

These properties apply across UAV routing, logistics, distributed optimization, and analog hardware — the domains differ, but the mathematical principles are the same.

Stability

$e^{-\gamma t}$ shows exponential decay, with errors shrinking smoothly over time.

$$\|z(t) - z^*\| \leq Ce^{-\gamma t}$$

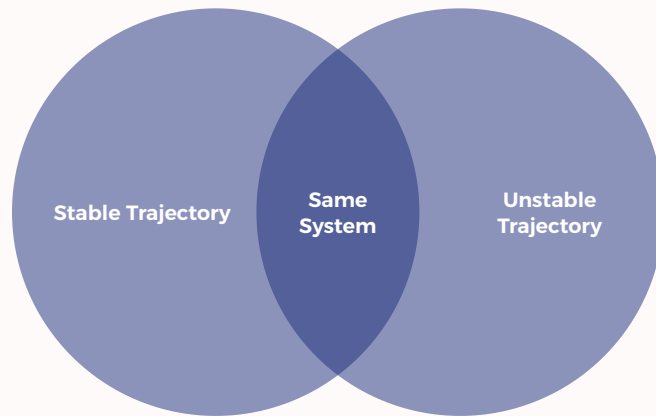
Anytime Usability

Useful intermediate iterates can be extracted at any time t before full convergence, so partial progress can still inform decisions.

Future Research Directions

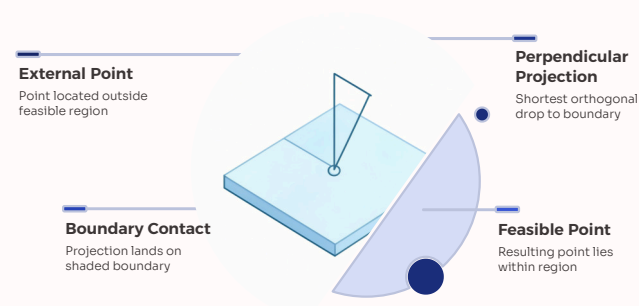
Implicit Integrators

Methods like backward Euler handle stiffness better and maintain stability.



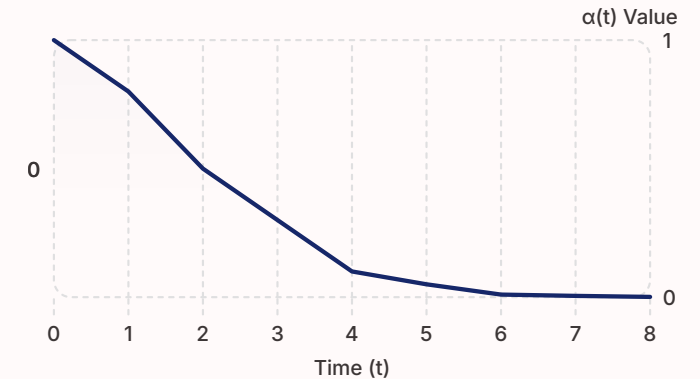
Projected Flows

¹ Adding projection steps keeps solutions feasible, related to mirror descent.



Adaptive Regularization

² Time-varying $\alpha(t)$ keeps early progress stable while reaching true optimum.

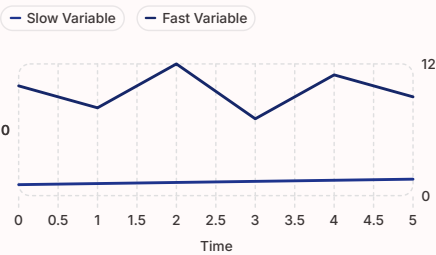


Current Limitations

While regularization brings significant benefits, it's important to acknowledge and address the inherent challenges that arise in practice. Understanding these limitations is key to developing more robust and efficient optimization systems.

Stiffness Mismatch

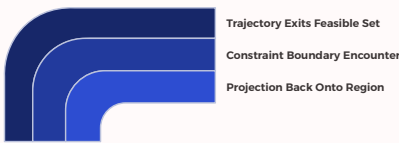
Variables change at very different speeds, causing simulation instability.



The graph illustrates how one variable can oscillate rapidly while another changes slowly, posing numerical challenges.

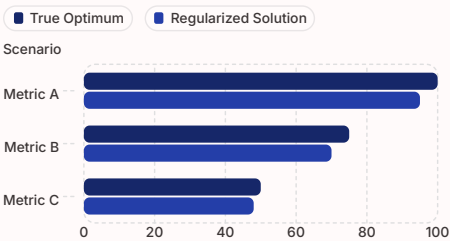
Boundary Violations

Continuous flows can briefly break constraints, requiring projection methods.



Regularization Bias

Regularized solution differs from true optimum, gap depends on α .



This bar chart compares the true optimum value for several metrics against the values obtained from a regularized solution, showing the inherent bias.